

Addressing Table & Device Configuration

Kestrel Industries

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Introduction

This document is the “who’s who” of the Kestrel Industries network. I’ve mapped out exactly which IP goes where, from the router interfaces to the individual PCs and servers. Following the brief, I’ve assigned the first usable address in each subnet as the default gateway for the LANs and the last usable address for the static end devices. This makes the network much easier to manage and troubleshoot since there’s a clear pattern to everything.

The Addressing Table

Here’s the full list of devices and their assigned addresses.

Device	Interface	IP Address	Subnet Mask	Default Gateway
Admin	G0/0 (Finance)	10.20.5.113	255.255.255.248	N/A
	G0/1 (Proc.)	10.20.5.49	255.255.255.240	N/A
	G0/2 (Sales)	10.20.5.129	255.255.255.248	N/A
	Vlan 144 (WAN)	10.20.5.145	255.255.255.252	N/A
Core	G0/1 (IT)	10.20.5.97	255.255.255.240	N/A
	Vlan 144 (WAN)	10.20.5.146	255.255.255.252	N/A
	Vlan 148 (WAN)	10.20.5.150	255.255.255.252	N/A
	S0/3/0 (WAN)	10.20.5.153	255.255.255.252	N/A
	S0/3/1 (WAN)	10.20.5.157	255.255.255.252	N/A

Ops	G0/0 (HR)	10.20.5.33	255.255.255.240	N/A
	G0/1 (Exec.)	10.20.5.121	255.255.255.248	N/A
	Vlan 148 (WAN)	10.20.5.149	255.255.255.252	N/A
West	G0/0 (Cust.A)	10.20.5.1	255.255.255.224	N/A
	G0/1 (Cust.B)	10.20.5.81	255.255.255.240	N/A
	S0/3/0 (WAN)	10.20.5.154	255.255.255.252	N/A
	S0/3/1 (WAN)	10.20.5.161	255.255.255.252	N/A
East	G0/0 (Eng.)	10.20.5.65	255.255.255.240	N/A
	G0/1 (R&D)	10.20.5.137	255.255.255.248	N/A
	S0/3/0 (WAN)	10.20.5.158	255.255.255.252	N/A
	S0/3/1 (WAN)	10.20.5.162	255.255.255.252	N/A
DNS Server	NIC	10.20.5.98	255.255.255.240	10.20.5.97
Web Server	NIC	10.20.5.99	255.255.255.240	10.20.5.97
PC Finance	NIC	10.20.5.118	255.255.255.248	10.20.5.113

PC Proc.	NIC	10.20.5.62	255.255.255.240	10.20.5.49
PC Sales	NIC	10.20.5.134	255.255.255.248	10.20.5.129
PC IT	NIC	10.20.5.110	255.255.255.240	10.20.5.97
PC HR	NIC	10.20.5.46	255.255.255.240	10.20.5.33
PC Exec.	NIC	10.20.5.126	255.255.255.248	10.20.5.121
PC Eng.	NIC	10.20.5.78	255.255.255.240	10.20.5.65
PC R&D	NIC	10.20.5.142	255.255.255.248	10.20.5.137
Customer PCs	NIC	DHCP	(Dynamic)	(See Notes)

Configuration Highlights

Static Addressing & Servers

I've assigned the static addresses as requested. For the IT network (10.20.5.96/28), the DNS Server is on 10.20.5.98 (2nd available) and the Web Server is on 10.20.5.99 (3rd available). The Web Server at www.site15186.com is set up with a custom page saying: "Hello! I am student 15186".

Dynamic Addressing (DHCP)

The West router is doing the heavy lifting for the Customer networks. It's running a DHCP server that automatically gives out IPs to any PC that plugs in.

Customer A: PCs get their info from the 10.20.5.0/27 pool (Gateway: 10.20.5.1).

Customer B: PCs get their info from the 10.20.5.80/28 pool (Gateway: 10.20.5.81).

Final Conclusion

The network is fully addressed and ready for action. Every static device is sitting on its last usable address, the gateways are all on the first usable address, and the DHCP pools are active and ready for the customers. Connectivity is rock solid across all segments.

End of Report